

Supplementary information to

White matter volume and white/gray matter ratio in different mammalian species as a consequence of the universal scaling of cortical folding

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Here is a full description of what we propose to be the underlying mechanism of cortical folding [7], and how it leads to the volumes of V_G , V_T and therefore V_W .

The developing cerebral cortex consists of a budding gray matter with cell bodies of newborn neurons, and white matter with their axons or invading axons. Both gray and white matter are subjected to a variety of sources of uneven forces as they develop. Our model builds on the physics of minimization of effective free energy of a growing surface, like the developing cerebral cortex, that is subject to heterogeneous bulk forces (such as growth-related push-pull forces, axonal tension, and hydrostatic and atmospheric pressures) [7]. Like Van Essen's inspirational tension-driven folding hypothesis [11], it acknowledges the role of connectivity in driving cortical folding, but unlike that model, it makes no attempt to identify the sources, the exact spatial distribution of those forces, or their impact on the exact placement of the folds. Rather, our model focuses on predicting simply the *degree* to which an adult or developing cortex of a certain combination of surface area and thickness (which together define volume) will be found to be folded.

We start by modeling axonal response to traction forces. This response has two components, operating on different time scales. Instantaneously, there is an elastic response, characterized by a Young modulus Y [28,29]. Like pulling apart the ends of a rubber band, an external longitudinal force F_{ext} induces an elongation $\Delta\ell$ in an axon that is balanced by the restorative force resulting from the deformation of the latter's cytoskeleton, so that $F_{ext} = Y \frac{\Delta\ell}{\ell}$.

It has been observed *in vitro* that, at longer time scales, a second, plastic response to traction forces F_{ext} ensues, altering the length of the axon [28]. This is characterized by an elongation rate $\frac{d\ell}{dt}$ that is proportional to the applied tension minus a threshold tension F_0 . The proportionality constant is an axon's sensitivity to tension $s(Y)$, that can in principle depend on the mechanical and geometrical characteristics of the axon. Neither s nor F_0 seem to depend on ℓ , although $s \sim \ell$ would also be plausible and would not change our results. This is mathematically represented by

$$\frac{d\ell}{dt} = s(Y, \ell) \left(\sum F_{ext} - F_0 \right) \quad (1)$$

Axonal elongation is probably the result of a signaling cascade triggered by instantaneous cytoskeleton deformation induced by external traction forces; the sensitivity s should thus be proportional to either $\Delta\ell$ or $\frac{\Delta\ell}{\ell}$ above the threshold. Writing F_0 and the external forces as gradients of potentials, we then obtain

$$\frac{d\ell}{dt} = -b(\ell)Y^{-1}\nabla(U_{ext} - F_0\ell) \quad (2)$$

This equation is mathematically equivalent to the damped 1-dimensional movement of a particle at position ℓ subject to an arbitrary outside potential, with an effective internal energy proportional to ℓ . Assuming the cortex changes slowly enough so that at each given time it is approximately in mechanical equilibrium, such ‘particle’ will attain a configuration that minimizes the term inside the parentheses, an effective potential energy. This corresponds to a minimization of the axon’s effective internal energy subject to external constraints. We assume Y and F_0 to be both proportional to axonal cross-sectional area a (this would also ensure a bundle of axons of different diameters under the same tension will elongate at the same rate, and account for observed different response rates observed for different axons). Furthermore, *in vivo* the external forces acting on the axon must result from tensions from within the grey matter it connects to. Thus we write the associated potential term as the surface tension energy of a grey matter layer growing under mechanical stress (in the simplest case, $U_{ext} = \sigma T(A_T - A_0(t))^2$, but in any case not explicitly dependent on A_E). In this case, for variations in the parameter A_E , the energy-minimization configuration will minimize the total white matter volume $V_W \propto \sum \ell_i a_i$. The proportionality constant will be an effective (negative) pressure, and the effective volumetric energy will be

$$\begin{aligned} E_V &= (p + p_{cs})V_W + \sigma T(A_T - A_0(t))^2 \\ E_V &= (p + p_{cs})(V - TA_T) + \sigma T(A_T - A_0(t))^2 \end{aligned} \quad (3)$$

where p is the effective pressure, p_{cs} is the cerebrospinal fluid pressure, V_w is the white matter volume, V is total cortical volume, T is the average cortical thickness and A_T is the total cortical surface area.

In a configuration ensemble, it is clear that the effective energy above is trivially minimized when V_w goes to 0. But the smaller the volume, the greater the fraction of configuration corresponding to non-physical self-intersecting surfaces. We thus add to the energy a term that penalizes such intersections so that their contribution gets suppressed exponentially in an effective free energy.

$$E_s[\vec{s}, n] = \int d^2\vec{x}_1 d^2\vec{x}_2 \delta^2(\vec{s}(\vec{x}_1) - \vec{s}(\vec{x}_2)) \Lambda_T(n(\vec{x}_1) - n(\vec{x}_2)) \Lambda_T \quad (4)$$

Above, $\vec{x}$ is the internal 2D coordinate of a point in the surface, and $\{\vec{s}(\vec{x}), n(\vec{x})\}$ is its image in $\mathbb{R}^3$, divided respectively into normal and tangential components.; $\Lambda_T(u) = T - 2|u|$ if $-\frac{T}{2} < u < \frac{T}{2}$ and 0 otherwise.

This last term is very awkward, so we approximate it by using a so-called Flory [30] approximation, where we substitute the integral by an estimate of the number of potential ‘collisions’ between patches of grey matter for each macro-state. Nelson et al. [31] is the standard reference for the statistical physics of surfaces (see [31], chapters 5.3.3 and 6.3, and also [32]), but Flory approximations are more heuristic than proof. In any case, we assume that the effective free energy scales as T^3 once express the areas in dimensionless units $\frac{A_T}{T^2}$ and $\frac{A_E}{T^2}$, while the rate of ‘collisions’ should be proportional to the square of the ‘concentration’ $\frac{A_T}{A_E}$ (the gyrification index!). We thus obtain

$$E_s \propto T \left(\frac{A_T}{A_E} \right)^2 A_E \quad (5)$$

The effective free energy F is $\propto E_v + E_s$, and the equilibrium configuration is the one in which it is minimized, $\frac{\partial F}{\partial A_E} = 0$, and thus

$$\begin{aligned} T^{1/2} A_T &\propto (p + p_{cs}) A_E^{5/4}, \\ T^{1/2} A_T &= k A_E^{5/4} \end{aligned} \quad (6)$$

We can rewrite equation (6), which relates areas and the cortical thickness, in terms of the volumes. Given that the grey matter volume is $V_G = A_T T$, and assuming the relationship between volume and area in cortical hemispheres is the same as the one found in geometrical hemispheres, we get

$$V_T = \frac{2}{9\sqrt{3}\pi} A_E^{5/2} = 0.0724 A_E^{1.5} \quad (7)$$

one obtains a power law relation between the grey matter and total volumes

$$V_g T^{-1/2} = k \left(\frac{9\sqrt{3}\pi}{2} V_T \right)^{5/6} \quad (8)$$

By replacing V_T with $V_g + V_w$, we obtain an explicit relation between the volumes of white and grey matter:

$$V_w = \frac{2}{9\sqrt{3}\pi} k^{-6/5} V_g^{6/5} T^{-3/5} - V_g \quad (9a)$$

This is not a power law, even if it appears to behave as one for the interval of values found in mammalian species. Still, this is the equation that does relate V_w to V_g universally across mammalian clades, with an added dependence on T . Like all previous relations derived from Eq. (1), it can be rewritten in terms of dimensionless quantities where T defines a fundamental length scale; in this case $v_w = \frac{V_w}{T^3}$ and $v_g = \frac{V_g}{T^3}$, and

$$v_w = \frac{2}{9\sqrt{3}\pi} k^{-6/5} v_g^{6/5} - v_g \quad (9b)$$

Empirically, across different mammalian species, what we have is close to equation (6)

$$T^{1/2} A_T = k A_E^{1.302} \quad (10)$$

Because $A_T = V_G/T$, and, in accordance with (7), $A_E \sim V_T^{0.668}$, equation (10) can be rewritten as $V_G/T \cdot T^{1/2} \sim V_T^{0.668 \times 1.302}$ or $V_G \cdot T^{-1/2} \sim V_T^{0.870}$. Empirically, we find that $V_G \cdot T^{-1/2} \sim V_T^{0.872 \pm 0.006}$ ($r^2 = 0.998$, $p < 0.0001$),

a function that is in very good agreement with the prediction and applies universally across all clades.

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